

Atomically precise catalysts design toward efficient waste plastic hydrogenolysis

Received: 18 December 2025

Accepted: 27 May 2026

Published online: 06 June 2026



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Hydrogenolysis of waste plastics offers a promising route to valorize them into fuels, light olefins, and aromatics (BTX). Despite extensive studies, rational catalyst design and process optimization remain challenging. Although conventional catalysts (homogeneous, nanoparticle, bulk) have driven progress through interfacial and ensemble effects, the activity and stability of these systems ultimately originate from elementary steps at the atomic scale motivating a shift toward atomically precise active-site design. This Perspective highlights atomic active centers and key mechanistic factors and underscores the importance of techno-economic and life cycle analyses for atomically precise catalysts, providing guidance for designing efficient hydrogenolysis systems.

Plastics, as widely used synthetic polymers, possess high stability and durability but have led to severe environmental accumulation, with over 8.3 billion metric tons produced since the 1950s and more than 75% persisting in the environment^{1,2}. Current waste management strategies including mechanical, chemical, and biological recycling remain limited. Mechanical recycling suffers from material degradation and accumulation of non-intentionally added substances (NIAS), while biological and emerging electro-/photocatalytic approaches are constrained by low efficiency, complex pretreatment, and limited feedstock compatibility (Fig. 1)^{3–5}. In contrast, thermocatalytic hydrogenolysis has emerged as a technologically mature and versatile route, capable of converting diverse plastic feedstocks into valuable chemicals⁶. Typically, plastics are composed predominantly of C–C and C–H bonds, and can be represented by the general formula C_nH_{2n} for polyolefins, or $C_nH_{2n}X_m$ for heteroatom-containing polymers, where X denotes heteroatoms such as O, N, S, and Cl incorporated into the polymer backbone or side groups. Through hydrogenolysis, these C–C bonds can be cleaved to generate short-chain alkanes, making this process a promising route for producing fuel-range hydrocarbons and

chemical feedstocks. Moreover, the high hydrogen content stored in polyolefin C–H bonds also enables a complementary valorization mode in which plastics serve as solid-state chemical hydrogen carriers, releasing H_2 through catalytic dehydrogenation or aqueous-phase reforming. Catalytic release of hydrogen from waste plastics thus provides a potential pathway for hydrogen recovery and energy recycling^{7,8}. Hydrogenolysis can also effectively process plastics containing chlorine and other heteroatom dopants, producing separable small molecules, or transforming NIAS to prevent the accumulation of heteroatoms that would otherwise lead to catalyst or system deactivation^{9–11}.

Waste plastic degradation faces three key challenges: heterogeneous feedstocks, broad product distributions, and limited process efficiency. Real-world plastics are typically complex mixtures, whereas most catalytic systems require well-defined substrates, making direct conversion difficult^{12,13}. In addition, uncontrolled reaction pathways often lead to wide product distributions, hindering selective production of high-value chemicals¹⁴. Improving reaction rates and throughput is therefore essential for scalable implementation. Conventional

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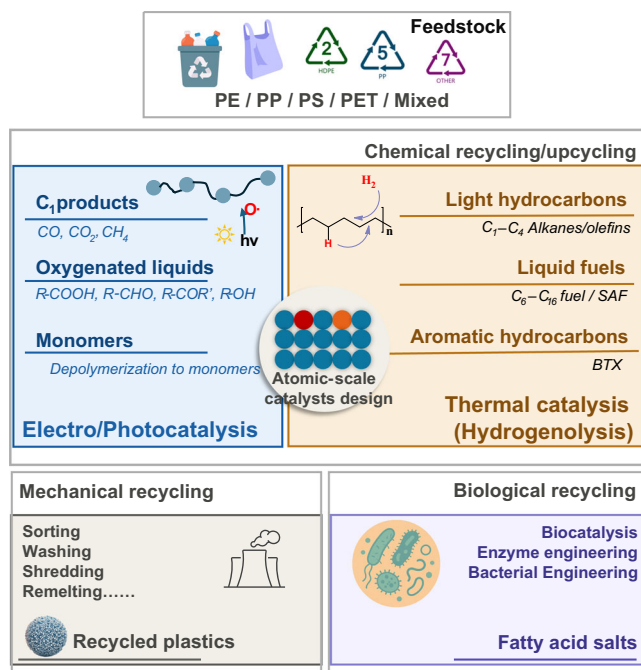


Fig. 1 | Catalyst-driven framework for plastic waste recycling and upcycling. Waste plastic treatment is divided into chemical recycling and upcycling, mechanical recycling, and biological recycling. For each category, the main processing routes and resulting products are presented, offering a product- and process-oriented perspective for identifying suitable valorization pathways for waste plastics.

bulk and nanoparticle catalysts, although industrially established, suffer from low atomic utilization, aggregation, and structural reconstruction under reaction conditions^{15–17}. Furthermore, structural heterogeneity leads to variations in catalytic behavior, which become particularly problematic in polymer hydrogenolysis due to complex reaction networks. Single-atom catalysts (SACs) offer a promising alternative by providing well-defined active sites, high atom utilization, and improved structural uniformity¹⁸. Their simplified coordination environments enable more reliable mechanistic understanding and can narrow product distributions. In addition, advances in artificial intelligence (AI) and machine learning (ML) have accelerated atomically precise catalysts discovery and optimization. These features position atomically precise catalysts as attractive candidates for addressing catalytic hydrogenolysis of waste plastics (CHWP) challenges at both mechanistic and process levels.

Investigating the atomic-scale active centers underlying hydrogenolysis is essential for resolving how hydrogen activation (M–H formation, spillover) and C–C bond cleavage are coupled at a single active ensemble, since the two elementary steps share the same site and must be balanced rather than optimized in isolation. In this perspective, the interactions among plastic-derived intermediates, hydrogen species, and atomic-scale catalytic interfaces, and identify key factors governing activity and stability. Furthermore, we discuss strategies for the rational design and synthesis of atomically precise catalysts including single-atom catalysts (SACs), dual-atom catalysts (DACs), and supported molecular (single-site homogeneous) catalysts as discussed throughout this perspective-tailored to specific plastic hydrogenolysis scenarios and highlight which structure-performance relationships should be prioritized. Because the application of atomically precise catalysts to plastic upcycling remains at an early stage, this perspective focuses primarily on SACs as the most extensively studied subclass, while drawing on lessons from DACs, single-atom alloys, and supported molecular analogues to outline how future SAC-based plastic hydrogenolysis catalysts might be designed. Leveraging

insights from structure–activity correlations, together with DFT-derived descriptors, can significantly accelerate the screening and optimization of atomically precise catalysts. Finally, we explore how techno-economic analysis (TEA) and life cycle assessment (LCA) can be used to assess whether the integration of atomically precise catalysts into the plastic hydrogenolysis process will impact bulk chemical feedstocks in the future. This framework links atomic-scale catalyst design with process-level performance, providing guidance for future CHWP development.

Design principles for atomically precise catalysts

Since the concept was introduced in 2011, SACs have undergone more than a decade of rapid development, encompassing advances in synthesis, precise structural design, and the elucidation of structure–activity relationships across diverse catalytic processes¹⁹. Catalyst design plays a central role of CHWP²⁰. Different catalytic pathways impose distinct requirements, and rational design must therefore be guided by specific performance metrics and process demands. SACs with potential in CHWP can be broadly categorized into four classes based on their physicochemical properties: carbon-based supports, oxide supports, metallic substrates, and immobilized molecular catalysts. This classification establishes a general platform for catalyst design under different CHWP scenarios (Fig. 2). Based on this framework, four key criteria—scalability, structural precision, stability, and regenerability—should be balanced to guide the development of effective catalysts.

Carbon-supported SACs are typically synthesized via confinement methods²¹, polymer pyrolysis²², and defect-mediated atom trapping¹⁶. These materials are typically prepared at small scales, with metal stabilization relying on heteroatom doping (e.g., N-doped carbon) or intrinsic defects, resulting in relatively low metal loadings and heterogeneous coordination environments. Carbon supports generally behave as Lewis bases and lack intrinsic proton-donating capability; consequently, CHWP activity mainly originates from the single-atom active centers or coordinated heteroatom sites. Isolated metal atoms can promote H₂ activation or selectively adsorb functional groups within polymer chains, thereby weakening C–C bonds and facilitating bond scission. Oxide-supported atomically precise catalysts are commonly synthesized via impregnation or defect-assisted atom trapping, while mesoporous oxides and zeolitic materials enable confinement-based approaches. Cation vacancies and oxygen defects stabilize high-coordination single-atom sites, whereas impregnation typically yields low-coordination configurations. High coordination enhances atomic stability, while low coordination provides higher intrinsic activity. Oxide-supported SACs can be synthesized at larger scales, although metal loadings are often limited unless defect-generating agents are introduced. Compared to carbon supports, oxides exhibit more uniform coordination environments and tunable Lewis acidity/basicity, with some hydroxyl-rich oxides additionally providing Brønsted acidity. However, because many oxides can intrinsically catalyze C–C transformations under CHWP conditions, rigorous control experiments are required to identify the true role of single-atom sites. Metal-supported SACs, often referred to as single-atom alloy or dilute metal alloys, are prepared by adapting nanoalloy synthesis methods, where reducing the secondary metal concentration keeps isolated atomic dispersion, or by surface-enrichment strategies such as cation exchange and galvanic replacement. These systems enable high atomic utilization and scalable synthesis. Isolated heteroatoms are stabilized at metal nanoparticle surfaces through electronic matching and free-energy minimization, while intermetallic chemistry can further enhance stability via strong metal–metal bonding. Metallic supports such as Co, Ni, Fe, and Cu readily activate small molecules²³, and decoration with isolated Pt, Ru, Pd, or Rh atoms modulates CHWP kinetics and product selectivity through bond-recognition effects²⁴. Immobilized homogeneous catalysts represent a fourth category, in

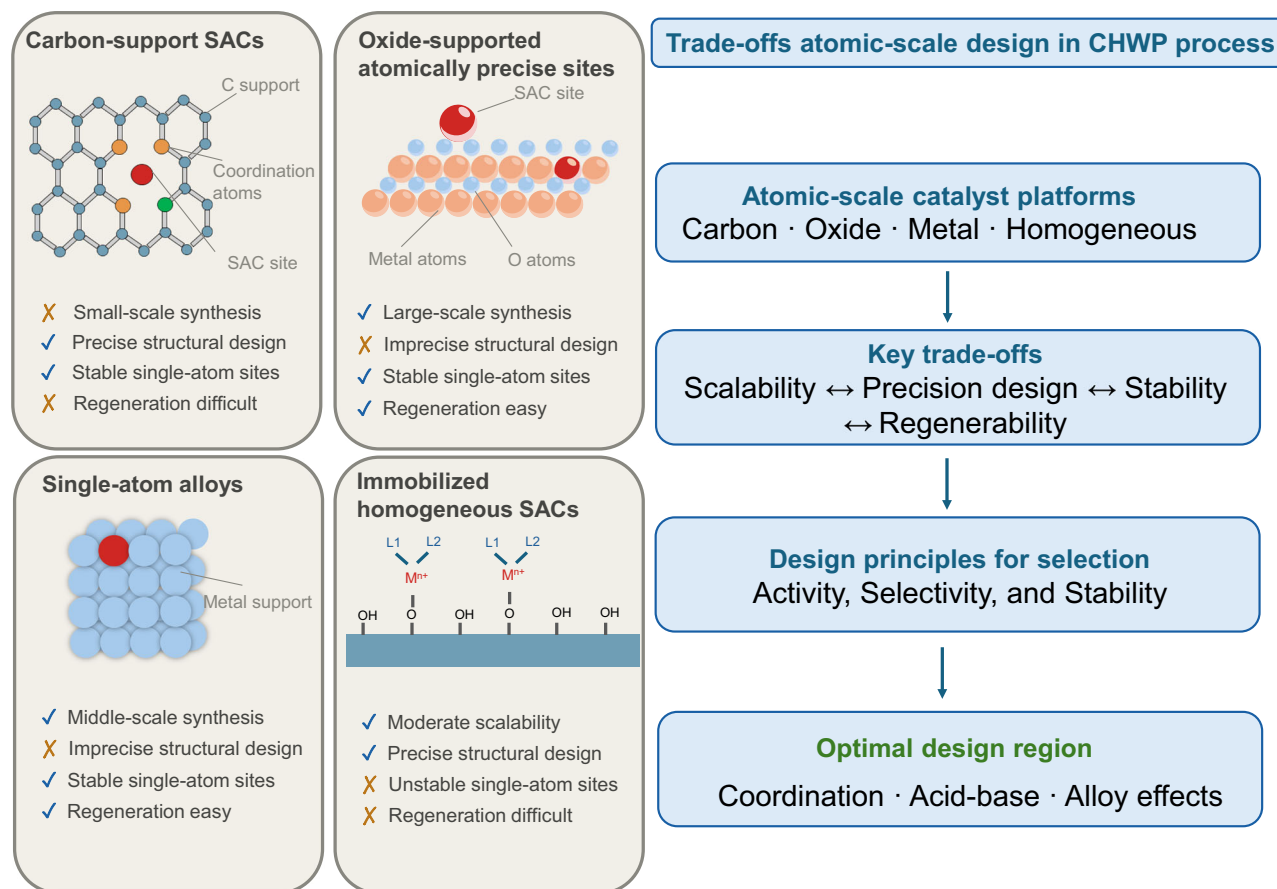


Fig. 2 | Trade-offs and design principles for atomic-scale catalyst design in plastic hydrogenolysis. Comparison of four representative platforms: carbon-supported single-atom catalysts, oxide-supported atomically precise sites, single-

atom alloys, and immobilized homogeneous catalysts. Across scalability, structural precision, stability, and regenerability, highlighting the trade-offs that guide catalyst selection for different CHWP scenarios.

which well-defined metal complexes are immobilized on catalytically inert supports such as silica or alumina via ligand exchange^{25–27}. This approach affords precisely defined coordination environments and enables atomic-level insight into metal-site dynamics. Although such systems can achieve high metal loadings and scalability, weak metal-support interactions often lead to structural evolution under harsh conditions. Nevertheless, under moderate reaction conditions, immobilized homogeneous SACs remain valuable platforms for elucidating fundamental structure–reactivity relationships relevant to CHWP catalyst design.

The performance of atomically precise catalysts in CHWP is governed by trade-offs among activity, selectivity, and stability, which are dictated by support-assisted intermediate handling, electronic tunability of active sites, and coordination-dependent resistance to deactivation. Accordingly, rational atomic-scale design therefore requires integrating these parameters to optimize catalyst performance for specific CHWP applications.

Active-sites design for catalytic activity

Inspired by homogeneous metal-organic catalysis, SACs are emerging as important candidates for CHWP due to their uniform active sites and high atomic utilization. More importantly, their synthetic accessibility, superior selectivity, and high stability position SACs as the next-generation replacements for nanocatalysts (Fig. 3a). SACs may fundamentally alter conventional reaction pathways established for nanoparticle catalysts, by selectively tuning activation energies of key elementary steps, thereby enabling more efficient plastic hydrogenolysis. In addition, SACs can be readily derived from conventional

catalyst synthesis routes at low metal loadings, while maintaining high activity and maximizing metal utilization.

In hydrogenolysis catalysis, atomically defined active sites contribute through three distinct functional roles (Fig. 3b): (1) activation of hydrogen molecules, generating hydrogen adsorption intermediates or free radicals that are subsequently transferred to plastic molecules; (2) accelerate identification and adsorption of specific functional groups in plastic chains and further breaking the C–C, C–O, and C–X (X = S, N, Cl...) bonds, facilitating addition and depolymerization reactions with hydrogen molecules; and (3) activation of C–H bonds in plastic molecules, enabling the transfer of endogenous hydrogen atoms or radicals to other C–C bonds and achieve depolymerization.

Hydrogen activation and spillover

Similar to classical homogeneous catalysts exemplified by Wilkinson's catalyst, Rh atomic centers can activate H₂ at the noble-metal center to generate metal–hydride (M–H) species. Heterogeneous noble-metal catalysts (Ru, Pd, and Rh) could also react with hydrogen molecules to form M–H intermediates, and then the activated H intermediates would transfer through the support or directly to plastic molecules, leading to C–C bond cleavage and depolymerization. At the atomic scale, hydrogen spillover phenomenon could make the M–H intermediate transfer to the support surface, increase catalytic efficiency and the active interface area of catalysts. Since its introduced by Khoobiar et al.²⁸, hydrogen spillover has been widely happened in hydrogenolysis catalysis, particularly in interactions between Pd, Pt, Ru, and Ni nanoparticles and metal oxides such as FeO, CoO, WO₃, and ZnO^{29–32}. In this process, noble metal nanoparticles dissociate

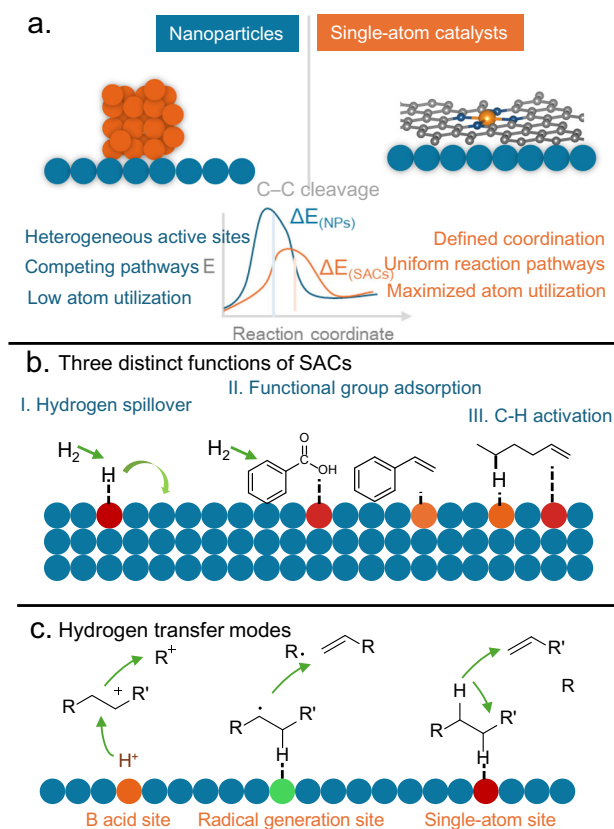


Fig. 3 | Atomic-scale mechanisms of hydrogen activation and C-C bond cleavage in plastic hydrogenolysis. **a** Comparison between nanoparticles and SACs, highlighting the influences of activity in CHWP process. **b** Three distinct functions of plastic molecular activation occurring at atomically defined active sites. **c** Alternative hydrogen transfer pathways at the atomic scale.

hydrogen to form Metal-H intermediates, while valence state exchanges between metal nanoparticles and oxygen in the support facilitate rapid $*H$ transfer to the catalyst surface, accelerating the catalytic reaction. Hydrogen spillovers can also occur on non-reducible oxides such as Al_2O_3 and zeolites, with entropy considerations influencing spillover activity. However, traditional hydrogen spillovers primarily enhance hydrogen activation on noble metal nanoparticles³³. With the development of single-atom alloys (SAAs), also known as dilute metal alloys, researchers have discovered that transition states such as Pd-H, Pt-H, and Ru-H can form at the active sites of dispersed noble metal atoms, which possess the ability to adsorb and dissociate hydrogen atoms before transferring them to the metal-support interface^{34–36}. In our recent work, we demonstrated that a La–Ru dual-atom catalyst supported on CoAl oxide enables efficient hydrogen transfer through electronic modulation between the two metal centers³⁷. Specifically, Ru–H species can be regulated by adjacent La sites, facilitating rapid transfer of activated hydrogen from Ru to the support, where it accumulates and redistributes to sustain a high density of active Ru sites. This process enhances the adsorption of key intermediates such as phenol during polyester depolymerization, thereby accelerating aromatic ring hydrogenation to produce cyclohexanol. CoO_x-based supports have also been widely recognized as effective platforms for promoting hydrogen spillover and are commonly employed in hydrogenolysis reactions of plastics such as PET³⁸. Unlike hydrogen and oxygen vacancy transfer in heterostructures, SAAs accelerate elemental hydrogen migration at the catalyst surface through electronic structure modifications of the support and active sites^{39,40}. Rational design of atomic-scale Ru, Pd,

and other hydrogen-activating sites thus offers an efficient pathway for C–C bond cleavage in CHWP.

Functional-group identification and bond activation

The adsorption of intermediates onto plastic molecules, followed by the activation and cleavage of C–C bonds, is a key mechanistic aspect of CHWP. Since the introduction of Ziegler–Natta catalysts in 1950, enabling industrial plastic production, Ti/Al metal sites have inspired catalytic design⁴¹. This has led to Al-containing ionic liquids such as $[C_4Py]Cl-AlCl_3$ ⁴², where Al centers are proposed to activate H_2 and recognize olefinic species through d– π interactions, facilitating intermediate adsorption and C–C bond cleavage to form short-chain olefins. Similarly, metal-organic homogeneous catalysts based on Ru, Mn, and Mo can activate H_2 and facilitate C–C cleavage via oxidative addition and reductive elimination. Additionally, through olefin recombination reactions, Re-, Ru-, Mn-, and Co-based homogeneous catalysts have been employed to generate C₃ olefins by adsorbing olefins and plastic intermediates in chain reduction processes^{43–46}.

SACs bridge homogeneous and heterogeneous catalysis, as isolated metal atoms can exhibit dynamic behaviors similar to homogeneous systems, including redox adsorption and addition–elimination processes, while remaining stabilized on supports (Fig. 3a). In hydrogenolysis, Ru and Pd nanoparticles stabilize unsaturated intermediates derived from plastic chains via $C=C/C=O$ adsorption, with the subsequent H-addition and bond cleavage steps proceeding via a Langmuir–Hinshelwood mechanism on the metal surface. Similarly, $Hf(OTf)_4$ has been employed as a cocatalyst, where the Hf metal centers can selectively recognize C–O bonds and promote their hydrogenolytic cleavage to form acids and olefins⁴⁷. Among triflate systems, Hf is particularly effective for C–O/C–H scission. Supported Hf species can function as auxiliary sites. Introducing Ce or Fe single atoms near active centers reduces surface C–C species coverage, lowering local charge density, suppressing over hydrogenolysis, and preventing methane formation^{48,49}.

However, excessive intermediate adsorption on nanoparticle catalysts can lead to catalyst surface coverage, blocking active sites and hindering further hydrogen activation and hydrogenolysis reactions. In a strict LH mechanism, productive turnover requires multiple co-existing surface sites to simultaneously accommodate substrate adsorption and dissociative H_2 activation; on a heavily covered nanoparticle surface, this dual demand is no longer satisfied. An Eley–Rideal (ER) or ER-like pathway, in which gas-phase H_2 reacts with the substrate-bound intermediate, therefore opens an alternative design axis for waste-plastic hydrogenation that has yet to be systematically explored. Our previous work shows that Ni and Co DACs stabilize benzene rings in polystyrene (PS) plastics⁵⁰ through cooperative adsorption between adjacent Co and Ni atomic sites, which simultaneously anchor the aromatic ring and the vinyl group and thereby redistribute the electron density of the vinyl moiety, modifying the C–C bond energy and influencing the hydrogenolysis. Because the Co and Ni metal centers used in that work retain intrinsic H_2 -activation activity, our proposed mechanism still invokes prior H_2 activation at the metal sites, followed by hydrogenation of the adsorbed intermediates. Whether the same chemistry can proceed through a genuine ER pathway on carbon-supported SACs where the carbon support itself contributes negligibly to H_2 activation remains an open mechanistic question that warrants further investigation. Beyond such spatial cooperation, atomic-scale ensembles also enable a second mode of synergy operating through the electronic structure of the active sites. For example, tandem catalysis involving Zr–O–Si and W–O–Zr active centers under alkaline conditions can generate negatively charged catalytic sites, which facilitate the formation of carbenium intermediates. This cooperative interaction leads to a catalytic performance that exceeds the sum of the individual components, exemplifying a true synergistic effect⁵¹. Moreover, the electronic

structure of isolated atoms allows precise tuning of d-band centers, optimizing intermediate adsorption and enhancing activity compared to nanoparticle systems.

Three approaches for H activation in CHWP

In conventional hydrocracking catalysis of polyolefin plastics, proton transfer facilitated by the Brønsted acid sites of acidic oxides, such as zeolites, catalyzes long chains with C–C bond cleavage into small saturated hydrocarbon gases, such as methane, via the Haag-Dessau mechanism, which involves the formation of carbenium ion intermediates (Fig. 3c)^{52,53}. Additionally, C–C bond cleavage can occur through the generation of H radicals and metal-H species, leading to unstable C intermediates. The utilization of three activated hydrogen species for efficient plastic scission further suggests that the controllable design of atomically dispersed metal-atom centers are the key determinants of CHWP performance.

Zeolites such as HZSM-5 and Y-type materials efficiently convert polyolefins into short-chain hydrocarbons via proton-mediated catalysis⁵⁴. However, relying on Brønsted acid site-driven hydrocracking via the Haag-Dessau mechanism tends to generate excessive short-chain alkanes, such as methane, while also requiring high reaction temperatures and significant energy input. Beyond intrinsically present Brønsted-acid sites, Ru single atoms supported on ZrO₂ are proposed to induce B-acid character at neighboring oxygen atoms through electronic charge transfer within Ru–O–Zr structure. These sites enable C–C bond cleavage via a carbenium ion-mediated mechanism⁵⁵. Compared to pathways dominated by metal-hydride or radical species, carbenium ion driven routes allow milder activation of internal C–C bonds in polyolefins, suppressing the formation of C₂ products and enabling selective production of jet-fuel-range hydrocarbons⁵⁵. Furthermore, the confinement effect of zeolites pores promotes arylation and alkane isomerization, leading to the formation of various by-products. Therefore, introducing additional hydrogen activation mechanisms is necessary for achieving selective product control. A common strategy involves the synergistic mechanism of metal-H species and carbenium ion intermediates, where noble metals such as Ru, Pt, Re, and Pd facilitate H–H bond cleavage in conjunction with zeolite catalysts, thereby improving product selectivity. However, the small pore size of zeolite presents challenges, as nanoparticles or charged metal-organic species can block the pores, limiting the accessible surface area and catalytic interfaces of CHWP⁵⁶. Anchoring single metal atoms within the micropores of zeolite can enhance the interaction between noble metal sites, H₂ molecules, and plastic substrates, thereby improving catalytic efficiency.

SACs can further regulate carbenium formation through different H₂ activation pathways, improving selectivity by enhancing activity and narrowing product distributions. Tuning the electronic density of Ru centers enhances the selectivity of polystyrene conversion to toluene by weakening hydrogen adsorption and suppressing over-hydrogenolysis⁵⁷. Organo-zirconium catalysts with different ligands can induce stronger acidic sites on oxide supports, accelerating HDPE scission. These insights highlight that atomically precise catalysts design can leverage Brønsted-acidic sites and coordination-environment engineering to regulate metal-hydride formation and control C–C bond cleavage rates and selectivity^{58,59}. Ru–Ni alloys display strong synergistic effects in hydrogenolysis catalysis, where electronic-density modulation regulates the population of allylic C₃ radical intermediates during HDPE hydrogenolysis, suppressing terminal C–C cleavage and methane formation⁶⁰. Whereas Ru favors terminal scission and Ni promotes C₃ pathways, Ru–Ni atomic pairs balance kinetics and cleavage mechanisms to achieve improved selectivity. Fe supported on mildly acidic WO₃ nanosheets enables the conversion of polyolefins into aromatics via coupling and aromatization of C₂/C₄ radical intermediates, while Fe single atoms facilitate

cyclohexene dehydrogenation through electronic-density modulation⁴⁸.

Beyond the use of exogenous hydrogen sources, hydrogen within plastic molecules can also participate in C–H bond activation, leading to the formation of carbocations and subsequent C–C bond cleavage^{61,62}. These pathways also involve activated hydrogen species and can be catalyzed by metal-hydride-forming systems, and thus fall within similar design strategies. However, in a strict sense, the absence of an external hydrogen source means that such processes should not be classified as CHWP.

Active-site design for catalytic stability

CHWP involves the formation of numerous intermediate products, and the distinct heat and mass transfer properties of solid-liquid-gas phases can introduce unexpected challenges in continuous reactors. A major issue in catalytic hydrogenolysis is catalyst deactivation, which significantly impacts long-term stability. Javier et al. reviewed fourteen types of catalyst deactivation mechanisms⁶³, among which the most relevant to plastic hydrogenolysis include coke formation, active site migration and poisoning, and catalysts structural degradation (Fig. 4a–e).

Conventional catalyst design for plastic hydrogenolysis has focused primarily on nanoparticle-sized active sites and macroscopic features (pore architecture, three-phase interfacial transport, and particle-level mass-transfer) to handle the heterogeneity of real waste-plastic feedstocks. These macroscopic levers, however, leave the local coordination environment of each active center—the very feature that governs M–H formation, intermediate adsorption strength, and resistance to poisoning, sintering, and coke-induced site loss—largely unaddressed. Closing this gap requires moving the design effort from the particle to the atomic scale, where the elementary steps that control selectivity and stability actually live. Atomically precise catalysts offer a promising solution by providing uniformly dispersed active centers, reducing aggregation and deactivation while optimizing metal-support interactions to preserve active site integrity^{64,65}. The precise control of electronic structures and coordination environments at the atomic level enables fine-tuning of catalytic activity, minimizing undesired side reactions.

Suppressing coke formation

For polyolefin plastics such as PE, PP, and polyester plastics, condensation of incompletely cracked olefins during hydrocracking can lead to form coke (Fig. 4a). Carbon accumulation does not inherently imply permanent catalyst failure, as coked catalysts can often be regenerated and reused through appropriate treatment. Several mechanisms contribute to this phenomenon: in acidic mesoporous materials such as zeolites, the pore structure can hinder the transfer of liquid intermediates, resulting in coke that reduces the catalytic area of acidic sites, while aromatization products further exacerbate condensation reactions^{56,66–68}. Additionally, carbon accumulation is commonly observed on the surfaces of metal catalysts with non-microporous and mesoporous structures (e.g., Ni, Ru, Co, Pt, Re)^{69,70}, where carbenium ions or C radicals fail to undergo cleavage quickly enough, leading to rapid C–C bond coupling.

In our previous work, we synthesized Ru SACs supported on CoAl oxides, which exhibited no coking during the hydrogenolysis of mixed plastics (PP, PE, PET)⁷¹. This contrasts sharply with Ru nanoparticles, as the atomically dispersed active site structure of Ru significantly enhances the adsorption strength of plastic intermediates, while isolated Ru sites prevent the condensation of carbenium ions or radicals, allowing only complete hydrocracking to occur. Furthermore, the introduction of oxygen into the coordination environment of active sites can suppress the C–C coupling process, thereby reducing carbon accumulation on the catalytic interface. Isolated metal atom centers not only exhibit stronger substrate adsorption for plastic

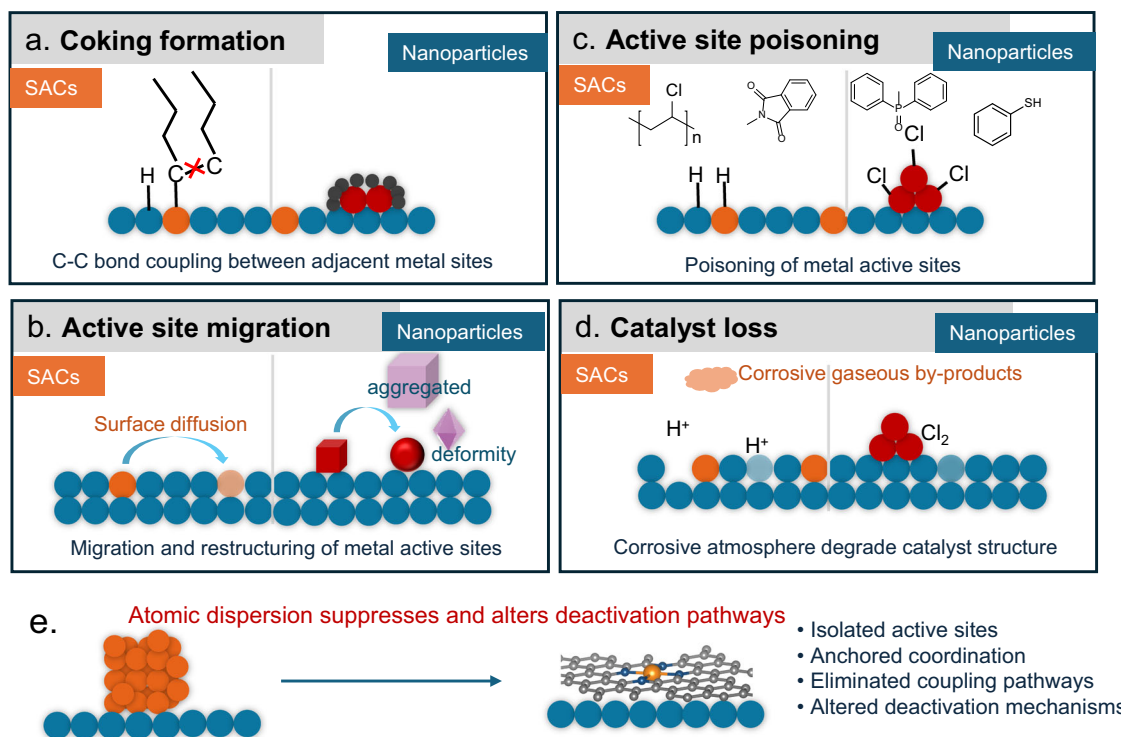


Fig. 4 | Atomic-scale deactivation mechanisms in CHWP. **a** Coking resistance; **b** active centers migration; **c** active centers poison; and **d** catalyst loss deactivation mechanism at the atomic level. **e** The key factor in enhancing the stability of the CHWP process with atomic-site design in atomically precise catalysts.

hydrogenolysis but also separate multiple C-containing intermediates, minimizing small molecular condensation. Additionally, carbon removal can be facilitated by lattice oxygen through suitable modulation. Another possible anti-coking mechanism is associated with the presence of strongly acidic sites (e.g. MoO_x) and the incorporation of abundant mesoporous structures⁷². While mesoporous materials offer large specific surface areas that provide sufficient active sites, they often suffer from limited substrate mass transport. In contrast, high-performance nanoscale atomically precise catalysts, despite possessing lower surface areas, can significantly enhance polymer mass transport and facilitate hydrogenolysis reactions, thereby mitigating coke formation. Since coking is widely recognized as a major cause of catalyst deactivation, the atomic-level design of plastic hydrogenolysis catalysts through isolated site effects and local coordination synergy presents a viable strategy for developing more stable anti-coking catalysts in the future.

Preventing active-site migration and poisoning

One of the most widely debated issues with SACs is the potential migratory reconfiguration of active sites (Fig. 4b), which can lead to stochastic structural shifts and result in single metal atoms that are not necessarily true active centers. Some studies suggest that single atoms undergo a catalytic cycle of migratory desorption-catalytic-return, while others propose that single atoms dynamically adapt their vibrational states under the influence of the ligand environment to achieve an optimal catalytic adsorption configuration and surface electronic structure.

In the hydrogenolysis of CHWP, nanoparticles of Ni, Ru, and Pt tend to agglomerate at high temperatures, in reducing atmospheres, and in the presence of complex intermediates, leading to the formation of large particles through sintering. Pt nanoparticles and Pt single atoms supported on activated carbon display markedly different selectivities toward liquid fuel hydrocarbons during HDPE hydrogenolysis, highlighting the critical role of active-site size and stability in

determining optimal performance⁷³. SACs also face challenges related to active site transition and reconfiguration during catalysis; however, the interaction between non-metal atoms (e.g., O, N, P, S) from the support and metal sites can inhibit metal atom migration and agglomeration. Unlike nanoparticles, which are primarily anchored to support via weak van der Waals forces, single metal atoms in SACs form strong covalent or coordination bonds with the support, significantly enhancing stability and resistance to agglomeration. This makes SACs a promising solution for mitigating active site deactivation caused by metal migration.

Beyond site reconfiguration, catalyst deactivation can also arise from elemental contamination, such as incomplete transfer of Cl, S, and N from waste plastics or CO poisoning during reactions (Fig. 4c)^{74,75}. In the case of chlorinated plastics like PVC, a tandem dechlorination-hydrogenolysis approach is commonly used for upgrading⁷⁶. Without pretreatment, chlorine-containing plastics generate HCl under H_2 atmospheres, which can severely damage catalyst structures. In contrast, AlCl_3 -containing ionic liquid catalysts can protect the Al active centers from HCl corrosion during hydrogenolysis cycles, enabling the stable catalytic processing of PVC into hydrocarbon and HCl products⁷⁷. Sulfur- and nitrogen-containing impurities in plastic feedstocks can generate strong Lewis bases (e.g., S^{2-} and NH_3) that poison Ni active sites¹⁰. The extent of deactivation strongly depends on metal-support interactions, and appropriately designed zeolitic supports, such as FAU, can significantly suppress such poisoning, highlighting the importance of support-single atom interactions in enabling poison-resistant catalysis. However, optimizing the coordination environment of SACs can improve impurity adsorption at active centers, reducing the risk of poisoning and extending catalyst lifetime^{78,79}.

Mitigating structural degradation

Structural loss of the catalyst is another major factor affecting stability in the CHWP (Fig. 4d). In hydrogenolysis reactions with batch reactors,

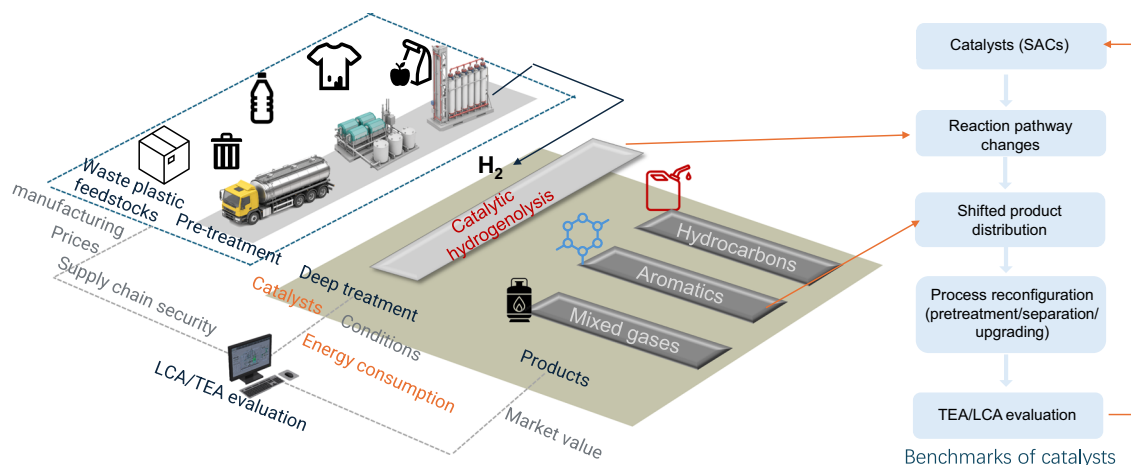


Fig. 5 | Process-level integration of atomically precise catalysts in plastic hydrogenolysis. Overview of waste plastics hydrogenolysis, linking catalyst design with reaction pathways, product distribution, and TEA/LCA-guided process optimization.

nanoparticles can dissolve under high-pressure and solvent-rich conditions, leading to catalyst degradation. Additionally, during hydrogenolysis of plastics containing impurity elements such as Cl, S, and P, acid gases like HCl, H₂S, and PH₃ are generated, which can corrode Al sites in oxide supports or zeolites, ultimately causing framework collapse. These by-product gases also impose stringent requirements on reactor materials, necessitating modifications to traditional piping and processing systems.

Catalyst stability is a critical parameter in plastic hydrogenolysis, and discussions on deactivation mechanisms are shifting from macroscopic observations to atomic-level insights into active sites and adsorption intermediate behavior. Therefore, designing catalysts at the atomic scale is essential for addressing carbon resistance, active site migration and poisoning, and catalysts loss, which are common challenges in plastic hydrogenolysis reactions. Atomically precise catalysts offer distinct advantages in this regard, as their unique adsorption mechanisms and tunable electronic structures enable precise control over adsorbed species, providing catalytic stability beyond that of conventional nanoparticle-based catalysts.

Computationally guided design of atomically precise catalysts

Atomically precise catalysts provide well-defined active sites that are highly suitable for theoretical modeling. Density functional theory (DFT) and ab initio molecular dynamics (AIMD) enable the correlation between atomic structure and catalytic performance by evaluating intermediate adsorption and reaction energetics. Compared to nanoparticle catalysts, the structural uniformity of SACs simplifies mechanistic analysis and improves predictive reliability^{80–82}. Furthermore, large-molecule reactions can be screened by exploring the synergistic effects of single or multi-metal centers through cascade mechanisms.

Key descriptors governing SAC performance can be broadly categorized into electronic structure, adsorption behavior, and stability. These include the d-band characteristics of active sites, adsorption energies of reaction intermediates, and stability-related parameters such as formation and aggregation energies^{83,84}. Data-driven approaches further accelerate SAC design by enabling the screening of metal identity and coordination structures. Although current machine learning applications in plastics primarily focus on polymer design and enzymatic degradation^{85,86}, integrating computational descriptors with catalyst design remains underexplored. Establishing reaction-specific descriptors, such as M–H formation energy and metal–intermediate interactions, will be critical for advancing SAC-enabled hydrogenolysis.

Environmental and economic evaluation

The increasing diversity of waste plastic treatment pathways has led to a lack of standardized evaluation metrics. A key debate revolves around whether plastics should be upcycled or downcycled^{87,88}. Changes in activity, selectivity, and stability due to SACs can alter conventional process pathways, making it necessary to assess whether SAC-enabled depolymerization can replace bulk chemical production routes. LCA and TEA are essential tools for evaluating both the environmental impact (e.g., carbon footprint, global warming potential (GWP)) and economic viability (e.g., minimum selling price (MSP), payback period) of different processes. Sensitivity analysis can further identify the key drivers governing the process feasibility performance and quantify their influence on the outcomes^{89–91}. For CHWP process, the value of the obtained chemicals is generally higher than that of the waste plastic feedstock, positioning it as an upcycling approach. However, most products from the hydrogenolysis process are crude bulk chemicals rather than monomers, limiting their direct reintegration into the polymer supply chain. Factors such as energy input, catalyst–plastic interactions, and material flow determine LCA indicators. By constructing full life-cycle models based on detailed process deconstruction and mass–energy balances, the distribution of carbon emissions across the entire process can be quantitatively assessed, enabling prediction of process-level optimization introduced by atomically precise catalysts. Comparative analysis among different reaction pathways allows evaluation of both carbon-emission mitigation and energy-efficiency improvements. For example, Javier et al. analyzed HDPE hydrogenolysis in batch reactors to assess the impact of Ru–Ni alloy catalysts, establishing correlations between product chain length, carbon emissions, and net profit⁶⁰. Longer product carbon chains were associated with higher profitability and lower GWP, while different energy inputs (wind, solar, fossil) led to similar optimal chain lengths, providing guidance for catalyst selection. Alternative LCA approaches further benchmark waste-plastic hydrogenolysis against conventional petrochemical routes or compare different hydrogen sources, such as green versus grey hydrogen. Besides catalyst/reaction performance, a critical challenge associated with CHWP processes is the stability of the waste plastic supply chain and whether its scale can sustain continuous bulk chemical production (Fig. 5). Consequently, feasibility assessments in this domain place significant emphasis on the comprehensive design of the process route, including feedstock supply, pretreatment, catalytic hydrogenolysis, and downstream product distribution. Accordingly, future LCA research is expected to expand toward the full value chain, from upstream recovery of legacy landfill waste to downstream substitution of virgin fossil resources by CHWP products.

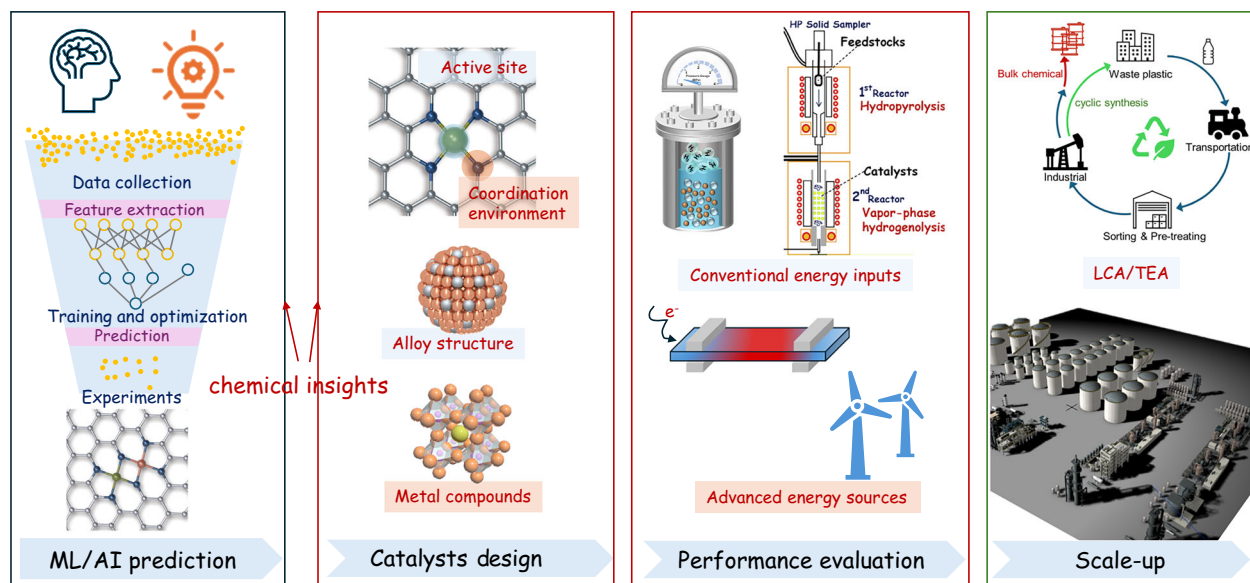


Fig. 6 | Development roadmap for atomically precise catalysts in plastic hydrogenolysis. Flowchart outlining the progression from proof-of-concept to industrial application, encompassing the prediction and understanding of

structure–performance relationships, the rational design of atomically precise catalysts, catalytic hydrogenation/hydrogenolysis driven by diverse energy sources, and the assessment of scale-up feasibility.

A more comprehensive, process-level metric is needed to validate whether the atomically precise catalysts-altered plastics upcycling process is likely to be a true effective route for the industry. Previously, the National Renewable Energy Laboratory (NREL) conducted LCA and TEA analyses on four catalytic conversion pathways for waste plastic: naphtha production, mixed-product formation, aromatics-rich production, and olefins-rich production⁹². These studies identified key economic factors influencing the process, including feedstock supply costs, operating expenses, catalyst costs, and separation and heating costs. The relative impact of these factors varies depending on geopolitical conditions. For example, in our work of waste polystyrene conversion to toluene over SACs using a tandem reactor configuration, TEA of both Operating Expenditure (OPEX) and Capital Expenditure (CAPEX) reveals that hydrogen price and catalyst cost are the dominant factors governing the MSP. Based on grey hydrogen pricing, the MSP of toluene derived from waste plastics is estimated to be as low as USD 0.61 kg^{−1}, which is significantly lower than the approximately USD 1.0 kg^{−1} associated with fossil-based production routes⁵⁷. Notably, hydrogen pricing is strongly influenced by policy incentives and upstream developments in hydrogen production technologies. As a major component of OPEX, catalysts therefore play a central role in determining the economic viability of catalytic waste plastic conversion processes. Currently, SACs constitute a major component of the MSP due to the absence of mature large-scale production methods and their high reference price. However, SACs exhibit high metal utilization efficiency and benefit from diverse synthesis strategies, which are expected to drive down costs in the future. Additionally, their homogeneous active sites and superior stability allow for better control over product selectivity, thereby reducing separation costs and extending catalyst lifetimes. Furthermore, their high catalytic activity can contribute to lower heating energy requirements. A holistic LCA approach reveals economic and environmental benefits of SACs that are not apparent when catalysts are evaluated in isolation. Most current LCA and TEA assessments rely on simulated process parameters that do not adequately capture the intrinsic influence of catalysts on reaction pathways. Sensitivity analysis can identify the dominant factors and link them to catalyst properties, enabling an assessment of how atomically engineered catalysts approach optimal process-level performance and providing quantitative guidance for catalyst design.

Future outlook

Hydrogenolysis is a particularly promising method, converting waste plastics into oils, fatty acids, short-chain hydrocarbons, light aromatics, and other valuable products. Scaling up plastic treatment requires systematic approaches, but multiple technological pathways exist. Among the variables that govern CHWP outcomes—feedstock composition, hydrogen source, reactor configuration, downstream separation—catalyst architecture is the most actionable lever, since it sets the achievable selectivity envelope and dictates the energy input and lifetime required by the overall process. Atomically precise catalysts with their well-defined active sites, high atomic utilization, and tunable coordination environments, have rapidly become an important research direction in chemical hydrogenolysis of plastics. Key factors in CHWP hydrogenolysis include plastic feedstock composition (single vs. mixed plastics)⁹³, hydrogen source (green, high-purity, low-purity), catalyst selection, reactor configuration (tandem, batch, continuous), and product categories (fuels, aromatics, gases, monomers, or carbon nano materials), all of which dictate the most effective treatment route. However, the feasibility of each approach requires rigorous validation. Future research must incorporate LCA and TEA to evaluate energy consumption, greenhouse gas emissions, and environmental impacts, as well as to optimize costs and benefits^{94,95}. These assessments will be crucial in determining the long-term feasibility of chemically catalyzed hydrogenolysis as a sustainable waste plastic valorization strategy (Fig. 6). From a process-level perspective, several directions merit further exploration. First, the rapid and efficient development of atomically precise catalysts requires the establishment of robust structure–performance relationships, supported by integrated platforms for synthesis, characterization, and catalytic evaluation. Second, leveraging material properties to accommodate more complex feedstocks and diverse hydrogen sources^{59,96} may enable broader and more flexible process design. Third, the development of new reaction configurations, guided by atomic-scale mechanistic understanding, can provide deeper insight into catalytic pathways. Finally, advances in operando characterization and process engineering are needed to standardize SAC preparation and enable scalable production. Together, these efforts will accelerate the industrial implementation of atomically precise catalysts in waste plastic hydrogenolysis.

The design of high-performance atomically precise catalysts can be understood in terms of activity and stability modulation in CHWP, guided by atomic-scale design based on existing mechanisms. The most critical step is C–C bond cleavage via exogenous or endogenous hydrogen. Whether through M–H intermediates, B–acid-induced carbocation formation, or radical pathways, understanding the atomic-level relationship between catalytic sites and reaction mechanisms is essential. In CHWP, both activity and stability are key factors in assessing catalyst viability at the microscopic level. SACs, due to their uniform and well-defined active sites, exhibit predictable dynamic behavior during reactions^{97,98}, making them highly tunable for catalytic performance optimization. AI and ML can further aid in identifying key parameters governing activity and stability, accelerating catalyst development. Although SACs significantly reduce the consumption of precious and active metals compared to traditional nanocatalysts, their high production cost remains a limiting factor for widespread adoption. However, as research progresses, continued advancements in cost-effective synthesis methods and scalable production techniques will further enhance the feasibility and affordability of SACs for CHWP hydrogenolysis.

Understanding structure–activity relationships in atomically precise catalysts enables more precise control of catalytic processes at the atomic level. Investigating the dynamic evolution of active sites and their physicochemical properties facilitates the identification of non-noble metal alternatives and optimization of coordination structures for improved adsorption/desorption and selectivity. In plastic hydrogenolysis, atomically precise catalysts behavior can be understood from three aspects: (1) the influence of electronic structures on plastic adsorption, (2) the role of atomic synergistic mechanisms in C–C bond cleavage, and (3) the activity of metal–hydrogen intermediates and hydrogen spillover on catalyst supports.

To enable efficient hydrogenolysis of waste plastics, the development of more affordable and accessible metal-based catalytic materials is urgently needed. Therefore, SACs hold great promise not only for their superior atom efficiency but also for their potential to achieve long-term catalytic stability. Importantly, while current plastic hydrogenolysis processes are already supported by mature engineering models, the integration of atomically precise catalysts could fundamentally alter product distributions, enabling higher selectivity toward more valuable chemical products. As plastic hydrogenolysis evolves from homogeneous systems to nanocatalysts and ultimately to single-atom catalysis, advances in atomic-level catalyst design will be key to unlocking the next generation of sustainable plastic upcycling technologies.

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Author contributions

Z.Z. conceived, developed, and edited the manuscript. S.W. contributed to the TEA/LCA part description. S. S., W. C. and Y. Y. Suggestions were made regarding the catalyst and the catalytic mechanism. J. W., W. Z., H. P. and D. W. contributed to the provision of ideas in the field and all the authors contributed the refinement of the paper.

Funding

This work was supported by the National Natural Science Foundation of China (52376195, J. W. and 22325101, D. W.).

Competing interests

The authors declare no competing interests.

Additional information

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Peer review information *Nature Communications* thanks Antonio Martín, Huayang Zhang and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

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